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Probability & Statistics

Bayes' Rule – For events A and B such that $P(B) > 0$, we have:

$$\boxed{P(A|B) = \frac{P(B|A)P(A)}{P(B)}}$$

Remark: we have $P(A \cap B) = P(A)P(B|A) = P(A|B)P(B)$.

Independence – Two events A and B are independent if and only if we have:

$$\boxed{P(A \cap B) = P(A)P(B)}$$

Variance – We define the variance of a random variable as follows.

$$\boxed{\mathrm{Var}(X) = E[(X - E[X])^2)] = E[X^2] - E[X]^2}$$

Standard deviation – The standard deviation σ of a distribution X is defined as:

$$\boxed{\sigma = \sqrt{\mathrm{Var}(X)}}$$

Covariance – More commonly known as $\mathrm{Cov}(X, Y)$.

$$\boxed{\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y}$$

Correlation – Let σ_X, σ_Y be the standard deviations of X and Y respectively. We define the correlation ρ_{XY}.

$$\boxed{\rho_{XY} = \frac{\sigma_{XY}^2}{\sigma_X \sigma_Y}}$$

Remark: If X and Y are independent, we have $\rho_{XY} = 0$. The converse may not be true.

KL divergence – The Kullback–Leibler divergence is a measure of how one probability distribution is different from a second, reference probability distribution.

$$\boxed{d(p_1(x), p_2(x)) = -\sum_x p_1(x) \log\left(\frac{p_2(x)}{p_1(x)}\right)}$$

Signal Processing

Convolution – Convolution is a mathematical operation on two functions (x and h) that produces a third function y expressing how the shape of one is modified by the other.

$$\boxed{y[n] = x[n] * h[n] = \sum x[m]h[n-m]}$$

Cross - correlation – Cross-correlation is a measure of similarity of two series as a function of the displacement of one relative to the other.

$$\boxed{(f \star g)[n] \triangleq \sum_{m=-\infty}^{\infty} \overline{f[m]} g[m+n]}$$

Discrete Fourier Transform – The discrete Fourier transform (DFT) converts a finite sequence of equally-spaced samples of a function into a same-length sequence of equally-spaced samples of the discrete-time Fourier transform (DTFT), which is a complex-valued function of frequency.

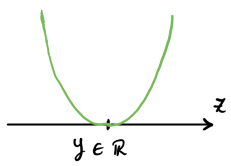
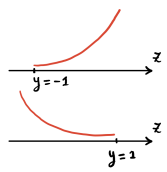
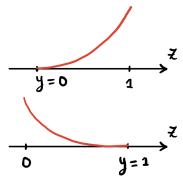
$$\boxed{X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-\frac{i2\pi}{N}kn}}$$

Short Time Fourier Transform – The short-time Fourier transform, is a Fourier-related transform used to determine the sinusoidal frequency and phase content of local sections of a signal as it changes over time.

$$\boxed{X[k, m] = \sum_{n=0}^{N} x[n]w[n - mH]e^{-j\frac{2\pi}{N}kn}}$$

Machine Learning

Loss function – Commonly denoted by L, this function quantifies how different our model prediction is from the actual outcomes. A few common loss functions are summed up in the following table:

Least squared	Logistic	Cross-entropy
$\frac{1}{2}(y - z)^2$	$\log(1 + \exp(-yz))$	$-[y \log(z) + (1 - y)\log(1 - z)]$
		
Linear regression	Logistic regression	Neural nets

Cost function – This measure is commonly used to assess the performance of a model. For a model h_θ (that we chose). It is defined as:

$$\boxed{J(\theta) = \sum_{i=1}^{m} L(h_\theta(x^{(i)}), y^{(i)})}$$

Gradient descent – Let α be the learning rate, the the update rule for our parameters θ using gradient descent is:

$$\boxed{\theta \leftarrow \theta - \alpha \nabla J(\theta)}$$

Likelihood – Used for finding optimal model parameters $\hat{\theta}$. In practice log-likelihood is used. $l(\theta) = \log(L(\theta))$.

$$\boxed{\hat{\theta} = \arg\max_\theta L(\theta)}$$

Normal equations – The value of θ that minimizes the cost function has a closed form solution, given by:

$$\boxed{\theta = (X^T X)^{-1} X^T y}$$

LMS algorithm – Let α be the learning rate, then the update rule for a training set of m data points, is as follows:

$$\boxed{\theta_j \leftarrow \theta_j + \alpha \sum_{i=1}^{m} \left[y^{(i)} - h_\theta(x^{(i)}) \right] x_j^{(i)}}$$

Logistic regression – We have the following form:

$$\boxed{\phi = p(y=1|x;\theta) = \frac{1}{1+\exp(-\theta^T x)} = g(\theta^T x)}$$

Softmax regression – We have the following form:

$$\boxed{\phi_i = \frac{\exp(\theta_i^T x)}{\sum_{j=1}^{K} \exp(\theta_j^T x)}}$$

Jensen's inequality – Let f be a convex function and X a random variable. We have the following inequality:

$$\boxed{E[f(X)] \geq f(E[X])}$$

Expectation-Maximization

Algorithm – E-step constructs a lower bound on the likelihood estimates and M-step optimizes that lower bound:

· E-step: Evaluate the posterior probability $Q_i(z^{(i)})$.

$$\boxed{Q_i(z^{(i)}) = P(z^{(i)}|x^{(i)};\theta)}$$

· M-step: Re-estimate cluster model using the posterior probability $Q_i(z^{(i)})$.

$$\boxed{\theta_i = \arg\max_\theta \sum_i \int_{z^{(i)}} Q_i(z^{(i)}) \log\left(\frac{P(x^{(i)}, z^{(i)};\theta)}{Q_i(z^{(i)})}\right)}$$

Principal component analysis

Algorithm – PCA reduces dimension by projecting data on k dimensions by maximizing the variance of the data.

· Step 1: Normalize the data to have 0 mean and unit variance.

$$\boxed{x_j^{(i)} \leftarrow \frac{x_j^{(i)} - \mu_j}{\sigma_j}} \text{ where } \boxed{\mu_j = \frac{1}{m}\sum_{i=1}^{m} x_j^{(i)}} \text{ and } \boxed{\sigma_j^2 = \frac{1}{m}\sum_{i=1} m(x_j^{(i)} - \mu_j)^2}$$

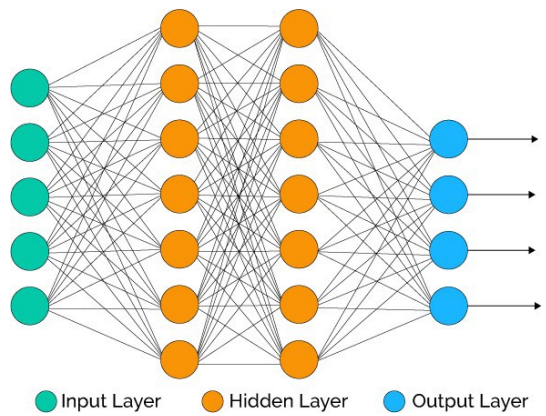
· Step 2: Compute $\Sigma = \frac{1}{m}\sum_{i=1}^{m} x^{(i)} x^{(i)^T} \in \mathbb{R}^{n\times n}$.

· Step 3: Compute $u_1 \ldots u_k \in \mathbb{R}^n$, the k largest eigenvectors of Σ.

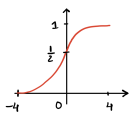
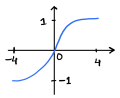
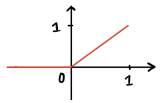
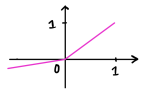
· Step 4: Project the data on to this space. This maximizes the variance among all k-dimensional spaces.

Neural Networks

Architecture – The vocabulary around neural nets architecture is described in the figure below:



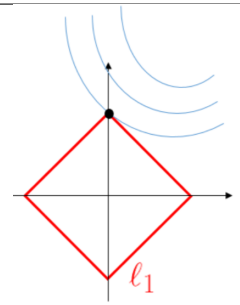
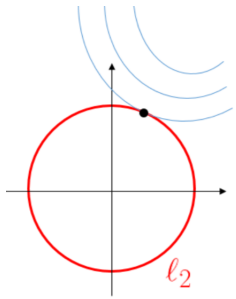
Activation function – These are used to introduce non-linear complexities at the end of a hidden unit.

Sigmoid	tanh	ReLU	Leaky ReLU
$g(z) = \frac{1}{1+e^{-z}}$	$g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	$g(z) = \max(0, z)$	$g(z) = \max(\alpha z, z)$ with $\alpha \ll 1$
			

Cross-entropy loss – In the context of neural networks, the cross-entropy loss $L(z, y)$ is commonly used. It is defined as:

$$\boxed{L(z, y) = -\left[y \log(z) + (1 - y) \log(1 - z)\right]}$$

Regularization – We perform regularization to avoid our model overfitting the training data.

LASSO	Ridge
Shrinks coefficients to 0 Good for variable selection	Makes coefficients smaller
	
$\cdots + \lambda \lvert\lvert\theta\rvert\rvert_1, \quad \lambda \in \mathbb{R}$	$\cdots + \lambda \lvert\lvert\theta\rvert\rvert_2^2, \quad \lambda \in \mathbb{R}$

Backpropagation – Backpropagation is a method to update the weights in the neural network by taking into account the actual output and the desired output. The derivative with respect to weight w is computed using chain rule and if of the following form:

$$\boxed{\frac{\partial L(z, y)}{\partial w} = \frac{\partial L(z, y)}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial w}}$$

As a result the weight is updated as follows:

$$\boxed{w \leftarrow w - \eta \frac{\partial L(z, y)}{\partial w}} \text{ where } \eta \text{ is the learning rate.}$$

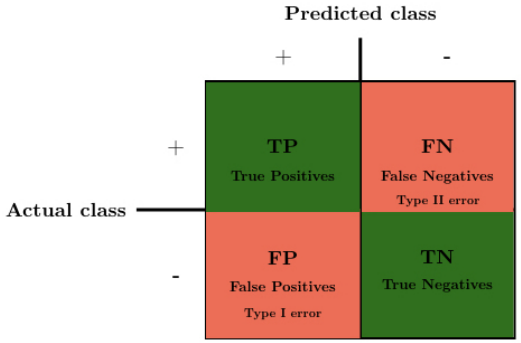
Updating weights – In a neural network, weights are updated as follows:

- Step 1: Take a batch of training data.
- Step 2: Perform forward to obtain corresponding loss.
- Step 3: Backpropagate the loss to get the gradients.
- Step 4: Use the gradients to update the weights of the network.

Dropout – Dropout is a technique meant to prevent overfitting the training data by dropping out neurons in a neural network.

Evaluation Metrics

Confusion Matrix – The confusion matrix is used to have a more complete picture of a model. It is defined as follows:



Main metrics – The following metrics are often used to measure the performance of a classification model:

Metric	Formula	Interpretation
Accuracy	$\frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}$	Overall Performance of the model
Precision	$\frac{\text{TP}}{\text{TP}+\text{FP}}$	Accuracy of positive predictions
Recall	$\frac{\text{TP}}{\text{TP}+\text{FN}}$	Coverage of actual positive samples
F1 score	$\frac{2\text{TP}}{2\text{TP}+\text{FP}+\text{FN}}$	Hybrid metric useful for unbalanced classes